Seminar 6 – GR 312CD. – 11.11.2021.

Ex 1 ? M_f^{Bc}; $\ker f$; $\operatorname{Im} f$.

a) $f: \mathbb{R}[X]_{\leq 4} \to \mathbb{R}[X]_{\leq 3}$; $f(P) = 2P' + 3P^4$.

b) $f: \mathbb{R}_{\leq 3}[X] \to \mathbb{R}^3$; $f(P) = (0, P(0)+2P(1), P'(1))$.

Ex 2 $f: \mathbb{R}^3 \to \mathbb{R}^3$; $f\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x+y \\ -x-y \\ z \end{pmatrix}$ a) $\operatorname{Im} f = ?$. b) $[f]_B = ?$

c) Cercetați dc $\exists$ o apl lin $g: \mathbb{R}^3 \to \mathbb{R}^3$ a.î. $g \circ f = id_{\mathbb{R}^3}$

$B = \left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}$.

Ex 3 V sp. vect; $\dim_k V = 7$; W_1, W_2 subsp în V.

$\dim W_1 = 4$; $\dim W_2 = 5 \Rightarrow W_1 \cap W_2 \neq \{0_V\}$

Ex 4 $f: \mathbb{R}^3 \to \mathbb{R}^2$; liniară; $f\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = f\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = f\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

a) $[f]_{Bc}$.

b) $\ker f$ c) ? $\exists g: \mathbb{R}^2 \to \mathbb{R}^3$ a.î. $g \circ f = id_{\mathbb{R}^3}$.

Ex 5 $f, g: V \to V$ lin a.î. $f \circ g = g \circ f$.

Ar. că $f(\ker g) \subseteq \ker g$; $f(\operatorname{Im} g) \subseteq \operatorname{Im} g$.

Ex 1 $f: \mathbb{R}[X]_{\leq 4} \to \mathbb{R}[X]_{\leq 3}$. $f(P) = 2P' + 3P''$

$B_c = \{1, X, X^2, X^3, X^4\}$ $B_c' = \{1, X, X^2, X^3\}$.

$f(1) = 2 \cdot 1' + 3 \cdot 1'' = \underline{0}$

$f(X) = 2 \cdot X' + 3 \cdot X'' = \underline{2}$

$f(X^2) = 2 \cdot (X^2)' + 3 \cdot (X^2)'' = 2 \cdot 2X + 3 \cdot 2 = \underline{4X+6}$.

$f(X^3) = 2 \cdot (X^3)' + 3 \cdot (X^3)'' = 2 \cdot 3X^2 + 3 \cdot 6X = \underline{6X^2 + 18X}$

$f(X^4) = 2 \cdot (X^4)' + 3 \cdot (X^4)'' = 2 \cdot \underline{4X^3} + 12X^2 \cdot 3 = 8X^3 + 36X^2$.

$$[f]_{Bc} = \left(\begin{array}{c|cccc} 0 & 2 & 6 & 0 & 0 \\ 0 & 0 & 4 & 18 & 0 \\ 0 & 0 & 0 & 6 & 36 \\ 0 & 0 & 0 & 0 & 8 \end{array}\right).$$

$\operatorname{rang} f = \operatorname{rang} [f]_{Bc} = 4$

$\dim(\operatorname{Im} f) = 4 = \dim \mathbb{R}[X]_{\leq 3} \Rightarrow$

$\operatorname{Im} f \subseteq \mathbb{R}[X]_{\leq 3}$;

$\Rightarrow \operatorname{Im} f = \mathbb{R}[X]_{\leq 3}$; $\{1, X, X^2, X^3\}$ bază pt $\operatorname{Im} f$.

$\ker f = \{P \mid 2P' + 3P'' = 0\} \subset \mathbb{R}[X]_{\leq 4}$.

$P(X) = a_0 + a_1 X + \dots + a_4 X^4$.

2. $P'(X) = a_1 + 2a_2 X + 3a_3 X^2 + 4a_4 X^3$.

3. $P''(X) = 2a_2 + 6a_3 X + 12a_4 X^2$. $\Rightarrow$

$0 = 2P' + 3P'' = (2a_1 + 6a_2) + (4a_2 + 18a_3)X + (6a_3 + 36a_4)X^2 + 8a_4 X^3$

$\Rightarrow a_4 = 0 = a_3 = a_2 = a_1$.

$\Rightarrow P(X) = a_0$. $\Rightarrow \ker f = \{a \mid a \in \mathbb{R}\} = Sp\{1\}$

$\dim \ker f = 1$.

Verificare $\dim \ker f + \dim \operatorname{Im} f = 5 = \dim \mathbb{R}[X]_{\leq 4}$. ✓

Ex2 $f: \mathbb{R}^3 \to \mathbb{R}^3$; $\operatorname{Im} f = ?$

a) $f\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x+y \\ -x-y \\ z \end{pmatrix}$; $\operatorname{Im} f = Sp\{f(e_1), f(e_2), f(e_3)\}$

$f(e_1) = f\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$; $f(e_2) = f\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$; $f(e_3) = f\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

$[f]_{B_c} = \left(\begin{array}{ccc} 1 & 1 & 0 \\ -1 & -1 & 0 \\ 0 & 0 & 1 \end{array}\right)$ $\operatorname{Im} f = Sp\left\{\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right\}$.

$\left\{\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right\}$ lin. ind $\Rightarrow$ bază pt $\operatorname{Im} f$.

$\boxed{\dim \operatorname{Im} f = 2}$

b) $[f]_B$; $B = \{\underbrace{\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}}_{v_1}, \underbrace{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}}_{v_2}, \underbrace{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}}_{v_3}\}$. $f\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x+y \\ -x-y \\ z \end{pmatrix}$.

$f(v_1) = f\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

$f(v_2) = f\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$; $f(v_3) = f\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix}$

$\Rightarrow [f]_{B,B_c} = \begin{pmatrix} 1 & 1 & 2 \\ -1 & -1 & -2 \\ 1 & 1 & 0 \end{pmatrix}$.

$[f]_{B,B} = ?$

$f(v_1) = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \alpha_1 \cdot v_1 + \alpha_2 v_2 + \alpha_3 v_3 = \alpha_1 \begin{pmatrix} 0 \\ +1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

$$\begin{cases} 1 = \alpha_2 + \alpha_3 & (1)-(3) \\ -1 = \alpha_1 + \alpha_3 \\ 1 = \alpha_1 + \alpha_2 \end{cases} \Rightarrow \boxed{\alpha_1 = \alpha_3} = -\frac{1}{2} \; ; \; \boxed{\alpha_2 = \frac{3}{2}}.$$

$$\Rightarrow [f(v_1)]_B = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} -1/2 \\ -1/2 \\ 3/2 \end{pmatrix} = [f(v_2)]_B .$$

$f(v_3) = \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix} = \beta_1 \cdot v_1 + \beta_2 v_2 + \beta_3 v_3 = \beta_1 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \beta_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \beta_3 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

$$\begin{cases} 2 = \beta_2 + \beta_3 \cdot] - \\ -2 = \beta_1 + \beta_3 \cdot \\ 0 = \beta_1 + \beta_2 \end{cases} \Rightarrow \begin{matrix} \beta_2 - \beta_1 = 4 \cdot] \pm \\ \beta_2 + \beta_1 = 0 \end{matrix} \Rightarrow \begin{matrix} \beta_2 = 2 \Rightarrow \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \\ 0 \end{pmatrix} \\ \beta_1 = -2 \cdot \\ \beta_3 = 0. \end{matrix}$$

$$[f]_B = \begin{pmatrix} -1/2 & -1/2 & -2 \\ -1/2 & -1/2 & 2 \\ 3/2 & 3/2 & 0 \end{pmatrix}.$$

c) ? $\exists g : \mathbb{R}^3 \to \mathbb{R}^3$ lin. $\tilde{i}$. $g \circ f = id_{\mathbb{R}^3}$.

$\dim \operatorname{Im} f = 2 \neq 3 = \dim \mathbb{R}^3 \Rightarrow f$ nu e surj.

$\dim \operatorname{Ker} f = \dim \mathbb{R}^3 - \dim \operatorname{Im} f = 1 \Rightarrow f$ nu e inj.

Prop. $\boxed{\begin{matrix} g \circ f = \text{surj} \Rightarrow g \text{ surj}. \\ g \circ f = \text{inj} \Rightarrow f \text{ inj}. \end{matrix}}$

Pp. $\exists g$ lin $\tilde{i}$. $g \circ f = \underbrace{id_{\mathbb{R}^3}}_{\text{inj}}$. $\Rightarrow f$ inj. Dar $\dim \operatorname{Ker} f = 1$. $\}$ ↯.

$\Rightarrow \nexists g$ lin $\tilde{i}$. $g \circ f = id_{\mathbb{R}^3}$. ▨

Ex 3 $\dim V = 7$; $\dim W_1 = 4$; $\dim W_2 = 5$; W_1, W_2 subsp în V.

$W_1; W_2 \subseteq V \Rightarrow W_1 + W_2 \subseteq V \Rightarrow \dim(W_1+W_2) \leq \dim V = 7$.

T. Grassmann: $\dim(W_1+W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$

$\Rightarrow \underbrace{\dim W_1}_{4} + \underbrace{\dim W_2}_{5} - \dim(W_1 \cap W_2) \leq 7 \cdot] \Rightarrow$

$\Rightarrow \dim(W_1 \cap W_2) \geq 2 \Rightarrow \dim(W_1 \cap W_2) \neq 0$.

$\Rightarrow W_1 \cap W_2 \neq \{0_V\}$. ☑

Ex 4: $f: \mathbb{R}^3 \to \mathbb{R}^2$; f lin.

$f\begin{pmatrix}0\\1\\1\end{pmatrix} = f\begin{pmatrix}1\\0\\1\end{pmatrix} = f\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}1\\1\end{pmatrix}$. $\Rightarrow$ a) $[f]_{B_C}$.

b) $\ker f$.

c) $\exists g$ lin $\cdot$ $g \circ f = id_{\mathbb{R}^2}$

$$f\begin{pmatrix}x\\y\\z\end{pmatrix} = f\left(x\begin{pmatrix}1\\0\\0\end{pmatrix} + y\begin{pmatrix}0\\1\\0\end{pmatrix} + z\cdot\begin{pmatrix}0\\0\\1\end{pmatrix}\right) \overset{f \text{ lin}}{=}$$

$$= x f\begin{pmatrix}1\\0\\0\end{pmatrix} + y \cdot f\begin{pmatrix}0\\1\\0\end{pmatrix} + z\, f\begin{pmatrix}0\\0\\1\end{pmatrix} = x f(e_1) + y f(e_2) + z f(e_3)$$

$f(e_1) = f\begin{pmatrix}1\\0\\0\end{pmatrix} = ?$

$$f\begin{pmatrix}0\\1\\1\end{pmatrix} + f\begin{pmatrix}1\\0\\1\end{pmatrix} + f\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}3\\3\end{pmatrix} \overset{f \text{ lin}}{=} f\begin{pmatrix}2\\2\\2\end{pmatrix} \Rightarrow f\begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}3/2\\3/2\end{pmatrix}.$$

$$f(e_1) = f\begin{pmatrix}1\\0\\0\end{pmatrix} = f\begin{pmatrix}1\\1\\1\end{pmatrix} - f\begin{pmatrix}0\\1\\1\end{pmatrix} = \begin{pmatrix}3/2\\3/2\end{pmatrix} - \begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1/2\\1/2\end{pmatrix}.$$

$$f(e_2) = f\begin{pmatrix}0\\1\\0\end{pmatrix} = f\begin{pmatrix}1\\1\\1\end{pmatrix} - f\begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}3/2\\3/2\end{pmatrix} - \begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1/2\\1/2\end{pmatrix}.$$

$$\boxed{[f]_{B_C} = \begin{pmatrix}1/2 & 1/2 & 1/2\\ 1/2 & 1/2 & 1/2\end{pmatrix}}.$$

$$f\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}1/2 & 1/2 & 1/2\\ 1/2 & 1/2 & 1/2\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}$$

b) $\ker f = ?$

$f\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{x+y+z}{2} \\ \frac{x+y+z}{2} \end{pmatrix} \Rightarrow \ker f = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} / \; x+y+z=0 \right\}$

$\ker f = \left\{ \begin{pmatrix} x \\ y \\ -x-y \end{pmatrix} \mid x, y \in \mathbb{R} \right\} = \mathrm{Sp}\left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right\}$

l. ind $\Rightarrow$ bază.

$\underline{\dim \ker f = 2}$.

c) $\nexists\, g: \mathbb{R}^2 \to \mathbb{R}^3$ lin i. $g \circ f = id_{\mathbb{R}^3}$.

$\mathbb{R}^3 \xrightarrow{f} \mathbb{R}^2 \downarrow g \searrow \mathbb{R}^3$

Pp $\exists g$ i. $g \circ f = \underbrace{id_{\mathbb{R}^3}}_{inj} \Rightarrow f$ inj.

$\dim \ker f = 2 \neq 0$ ↯ $\Rightarrow$ pp. a fost falsă

$\Rightarrow \nexists\, g: \mathbb{R}^2 \to \mathbb{R}^3$ lin i. $g \circ f = id_{\mathbb{R}^3}$.

Ex 5 $f, g: V \to V$ lin i. $f \circ g = g \circ f \Rightarrow$ $\underline{f(\ker g) \subseteq \ker g}$. $\underline{f(\mathrm{Im} g) \subseteq \mathrm{Im} g}$.

Trebuie să ar că $y \in \ker g \Leftrightarrow g(y) = 0$.

a) $y \in f(\ker g) \Rightarrow \exists x \in \ker g$ i. $y = f(x)$.

$\Downarrow g(x) = 0$ $\quad f$ liu

$g(y) = g(f(x)) = f(g(x)) = f(0) = 0$.

$\Rightarrow y \in \ker g \Rightarrow f(\ker g) \subseteq \ker g$.

b) $y \in f(\mathrm{Im} g) \Rightarrow \exists z \in \mathrm{Im} g$ i. $y = f(z)$.

$\Downarrow$

$\exists x$ i. $g(x) = z$.

$\Rightarrow y = f(g(x)) = g(f(x))$

$\Downarrow$

$y \in \mathrm{Im} g$.

$\Rightarrow f(\mathrm{Im} g) \subseteq \mathrm{Im} g$.

□

Valori și vectori proprii

$A \in M_n(K)$; $P_A(\lambda) = \det(A - \lambda I_n) = \begin{vmatrix} a_{11}-\lambda & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22}-\lambda & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn}-\lambda \end{vmatrix}$.

polinomul caracteristic asoc matr A.

$P_A(\lambda) = \left[\prod_{i=1}^{k} (\lambda - \lambda_i)^{n_i}\right] \cdot (-1)^n$ unde $\lambda_1, \dots, \lambda_k$ rād pol. cu ord de mult $n_1, \dots, n_k$.

$n_1 + n_2 + \dots + n_k = n$.

$\sigma(A) = \text{mult val proprii} = \{\lambda_1, \dots, \lambda_k\}$

$\vec{v} \in \mathbb{R}^n$ vector propriu pt val pr λ dc. $\boxed{A \cdot \vec{v} = \lambda \cdot \vec{v}}$ $\boxed{\vec{v} \neq \vec{0}}$.

$V_\lambda = \{\vec{v} / A\vec{v} = \lambda \cdot \vec{v}\}$ subsp vectorilor proprii asoc lui λ.

↓ sp. vect. ; $\boxed{\dim V_\lambda = g(\lambda) = \text{mult geom}}$.

Ex 6 $A = \begin{pmatrix} 1 & 0 \\ 5 & 2 \end{pmatrix}$; a) val proprii și vect pr pt A ⇒ $\lambda_1 = 1; \lambda_2 = 2$

b) — " — pt A^2 ⇒ $\lambda_1 = 1; \lambda_2 = 4$ cu aceiași vect pr

c) — " — pt A^{-1} ⇒ $\lambda_1 = 1; \lambda_2 = \frac{1}{2} = 2^{-1}$

Răspunsuri

a) $P_A(\lambda) = \begin{vmatrix} 1-\lambda & 0 \\ 5 & 2-\lambda \end{vmatrix} = (1-\lambda)(2-\lambda) \Rightarrow \lambda_1 = 1;\ n_1 = 1 = a(\lambda_1)$

$\lambda_2 = 2;\ n_2 = 1 = a(\lambda_2)$.

$\lambda_1 = 1$

$? \vec{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ i. $A\vec{v} = \lambda_1 \cdot \vec{v} \Leftrightarrow \begin{pmatrix} 1 & 0 \\ 5 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ 5x+2y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$.

$5x + 2y = y \Rightarrow y = -5x \Rightarrow \vec{v} = \begin{pmatrix} x \\ -5x \end{pmatrix} = x\begin{pmatrix} 1 \\ -5 \end{pmatrix}$. $\vec{v}_1 = \begin{pmatrix} 1 \\ -5 \end{pmatrix}$

$V_{\lambda_1} = \{\vec{v} / A\vec{v} = \lambda_1 \cdot \vec{v}\} = \text{sp}\{\vec{v}_1\}$; $\boxed{\dim V_{\lambda_1} = 1}$. ($\vec{v}_1$)

$\boxed{\lambda_2 = 2} \Rightarrow \vec{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ i. $A\vec{v} = \lambda_2 \cdot \vec{v} \Rightarrow \begin{pmatrix} x \\ 5x+2y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix} \Rightarrow x = 2x \Rightarrow \boxed{x = 0}$

$V_{\lambda_2} = \text{sp}\{\begin{pmatrix} 0 \\ 1 \end{pmatrix}\}$; $\dim V_{\lambda_2} = 1$ $\Rightarrow \vec{v} = \begin{pmatrix} 0 \\ y \end{pmatrix} = y\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Ex 7. $A \in M_n(\mathbb{R}) \Rightarrow \sigma(A) = \{\lambda_1, \dots, \lambda_k\}$.

a) Ar. că A inversabilă $\Leftrightarrow 0 \notin \sigma(A)$ (i.e. 0 nu e val pr)

b) Dc. A inversab $\Rightarrow P_A(\lambda) = 0 \Rightarrow P_{A^{-1}}(\lambda^{-1}) = 0$.

c) λ val pr $A \Rightarrow \lambda^k$ val pr pt A^k cu aceiași vectori proprii.

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a) ($\Rightarrow$) A inversab și pp. că 0 val pr $\Rightarrow$ $P_A(0) = 0$.

$P_A(\lambda) = \det(A - \lambda I_n) \Rightarrow P_A(0) = \det A$ } $\Rightarrow$

$\Rightarrow \det A = 0$ } fals. $\Rightarrow$ 0 nu e val pr.
A inversab

($\Leftarrow$) 0 val. pr. pt A. $\Rightarrow P_A(0) = 0 = \det A$ } fals.
Pp. A inversab $\Rightarrow \det A \neq 0$.

b) λ val pr pt $A \Rightarrow \lambda^{-1}$ val pr pt A^{-1}
A inversab $\Rightarrow \lambda \neq 0$

$P_{A^{-1}} \leftrightarrow P_A$.

$P_{A^{-1}}(\lambda) = \det(A^{-1} - \lambda \underbrace{I_n}_{A^{-1}A}) = \det\left(\underline{A^{-1}}\,(I_n - \lambda A)\right) =$

$= \det(A^{-1}) \cdot \det(I_n - \lambda A) = \frac{1}{\det A} \cdot \det\left[(-\lambda)\cdot\left(A - \frac{1}{\lambda} I_n\right)\right] \Rightarrow$

$A \cdot A^{-1} = I_n$.

$\det(A \cdot A^{-1}) = \det A \cdot \det A^{-1} = \det I_n = 1 \Rightarrow \boxed{\det A^{-1} = \frac{1}{\det A}}$.

$\Rightarrow P_{A^{-1}}(\lambda) = \frac{(-\lambda)^n}{\det A} \cdot \underbrace{\det\left(A - \frac{1}{\lambda} I_n\right)}_{P_A\left(\frac{1}{\lambda}\right)} = \frac{(-\lambda)^n}{\det A} \cdot P_A\left(\frac{1}{\lambda}\right)$. (*)

$\lambda \in \sigma(A^{-1}) \Rightarrow P_{A^{-1}}(\lambda) = 0 \overset{(*)}{\Rightarrow} P_A\left(\frac{1}{\lambda}\right) = 0 \Rightarrow \frac{1}{\lambda} \in \sigma(A)$.

$\lambda \neq 0$ (A inversab) ☐